



AI-Powered Intelligent Job Search and Career System

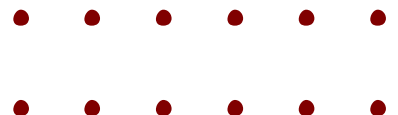
Data Science and AI Project

Milestone 5 Report

Group 4

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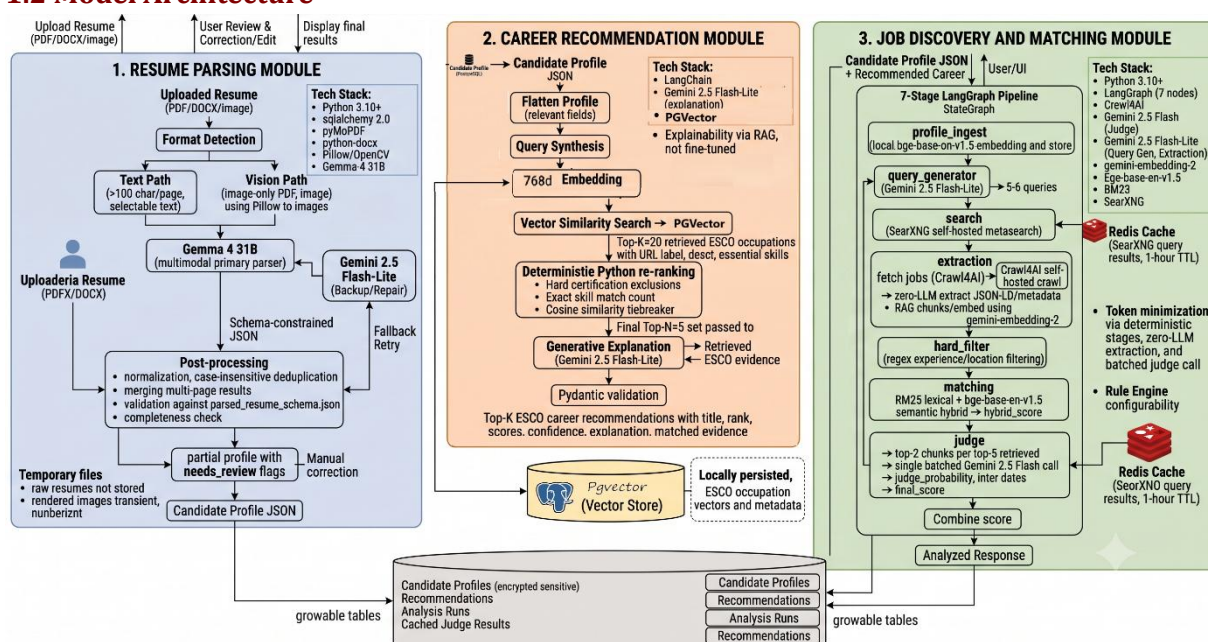
1. Introduction

1.1 Project objective and scope of this evaluation

The AI-Powered Intelligent Job Search and Career System converts an uploaded resume into a structured candidate profile, maps that profile onto occupations in a standard taxonomy, and uses the resulting role context to discover and rank live job postings. The three modules — Resume Parsing, Career Recommendation, and Job Discovery and Matching — communicate through a shared candidate profile and a shared database.

Milestone 5 is the evaluation and integration milestone, and it covers two kinds of work. Resume Parsing and Career Recommendation were merged into one FastAPI application over one centralised database, with the vector store migrated from a local index to hosted PostgreSQL with pgvector. Alongside that, Career Recommendation was evaluated on 86 annotated resumes, and Job Matching, Job Discovery, and the two prompt-driven stages were evaluated through a deterministic harness across four configuration sweeps. Job Discovery and Matching is evaluated at module level, since it is not yet wired into the shared profile flow.

1.2 Model Architecture



The models used at each inference stage are:

Module	Component	Model
Resume Parsing	Primary parser	Gemma 4 31B (multimodal, open-weight)
Resume Parsing	Backup and repair	Gemini 2.5 Flash-Lite
Career Recommendation	Embedding model	BAAI/bge-base-en-v1.5, 768 dimensions, local CPU
Career Recommendation	Vector store	Supabase PostgreSQL with pgvector, HNSW cosine index
Career Recommendation	Explanation model	Gemini 3.5 Flash-Lite
Job Discovery and Matching	Query generation	Gemini 2.5 Flash-Lite
Job Discovery and Matching	Judge	Gemma 4 31B
Job Discovery and Matching	Embedding model	BAAI/bge-base-en-v1.5, 768 dimensions, local CPU
Job Discovery and Matching	Lexical ranking	BM25 (rank_bm25)

1.3 Objectives of Milestone 5

- Merge Resume Parsing and Career Recommendation into one application over one database, and verify the merged path end to end on a real resume.
- Replace the local vector index with a centralised, hosted vector store shared across modules.
- Evaluate Career Recommendation against an annotated gold set with retrieval and ranking metrics, on both the full set and a held-out split.
- Evaluate Job Matching, Job Discovery, and the two prompt-driven stages through a deterministic harness, using configuration sweeps rather than single runs.
- Compare every selected configuration against an explicit baseline and record the failure cases that remain.

1.4 Nature of the work in this milestone

This system performs no gradient-based training. No weights are initialised, updated, or saved anywhere in the pipeline, because the project has no labelled training data of the kind these tasks would require. Several conventional headings, optimiser, learning rate, epochs, loss curves, checkpoint selection, therefore have no direct instance here. They are not skipped but mapped to its working equivalent in the configuration summary that follows.

2. Experimental Setup

2.1 Environment, frameworks, and libraries

Layer	Detail
Compute	No GPU anywhere. Gemma 4 31B and the Gemini-family models are called over hosted APIs; BAAI/bge-base-en-v1.5 (110M parameters) is the only local model and runs on CPU
Language and application	Python 3.10+ managed with uv; one FastAPI application exposing all module routers; SQLAlchemy 2.0 async with Alembic migrations
Storage	Supabase PostgreSQL, region ap-northeast-2, with pgvector; HNSW index using the cosine operator class; psycpg2 used for ingestion writes
Orchestration and ranking	LangChain components in Career Recommendation; a seven-node LangGraph StateGraph in Job Discovery; rank_bm25 for lexical scoring; sentence-transformers for embedding
Discovery services	SearXNG and Crawl4AI, both self-hosted, with a one-hour Redis result cache
Evaluation harness	promptfoo-based suite with rule-based Python validators; pytest for the test suite

All grading in the harness is deterministic and rule-based rather than model-graded. No llm-rubric assertion is used anywhere, because a model-graded assertion would cost a second model call per graded output and would make the evaluation itself non-reproducible.

2.2 Datasets, versions, and splits

Dataset	Version and size	Role and split
Resume corpus	Kaggle hadikp/resume-data-pdf: 8,905 single-page image-only PDFs	86-record annotated gold set, stratified two per category across 43 categories; development 34, held-out test 52, training 0

Dataset	Version and size	Role and split
Occupation taxonomy	ESCO v1.2.1 English: 3,039 occupations, 13,939 skills, 126,051 occupation-skill relations	Retrieval knowledge base, indexed in full. Not split — withholding occupations would degrade the system rather than create a held-out condition
Career Recommendation labels	134 category-to-occupation mappings across 43 categories	Ground truth for retrieval metrics; inherits the resume split
Job Matching golden set	86 candidates with a fixed 10-job pool each, 860 postings from the LinkedIn Job Postings Dataset	Ranking evaluation with real BM25 scores and real embeddings; candidate identifiers divided before pair construction at 60 / 20 / 20
Job Discovery and prompt sets	5 live search queries; 4 candidate profiles; 2 job-batch cases	Live search-and-crawl reliability and prompt sweeps, executed as real calls

Near-duplicate resume groups are confined to a single split, so a visually similar resume cannot straddle the evaluation boundary.

2.3 Reproducibility

- Random seed 42, fixed across split assignment and sampling; temperature 0 on every generative call in every module and sweep.
- Schema-constrained output on every generative call, validated before the output is accepted downstream.
- Model identifiers held in environment configuration rather than in source, so a retired model can be swapped without a code change.
- Database schema created by versioned Alembic revisions and a checked-in setup script, so the store can be rebuilt from scratch rather than reproduced by hand.

2.4 Evaluation workflow

Each measured component follows the same loop. The configuration surface is fixed; the ground truth and metric set are defined before the sweep runs, so the metric is not chosen after seeing the result; the component is run across a set of alternative configurations with everything else held constant; each configuration is scored by a deterministic validator and combined into one weighted composite by the aggregation script; the highest composite is selected automatically rather than by inspection; and the selected configuration is then re-run against an explicit baseline to confirm it earns its place.

3. System Configuration Summary

3.1 What replaces a training summary here

Because no weights are trained, the conventional training-summary headings are mapped onto their working equivalents, so that nothing in the standard reporting structure is silently dropped.

Conventional activity	Equivalent in this system
Trainable weights, optimiser, learning rate	None; no gradients are computed. The configurable surface prompts, schemas, retrieval parameters, and scoring weights, chosen by exhaustive sweeps over small discrete grids
Batch size	Survives only as a cost measure: the judge scores five shortlisted jobs in one call, and the parser sends one page image per call

Conventional activity	Equivalent in this system
Number of epochs	No repeated passes over training data. The occupation index is built in a single offline pass
Loss function	Deterministic rule-based validators combined into a weighted composite score, which is the quantity each sweep maximises
Early stopping	An explicit three-condition stopping criterion for prompt refinement, and a fixed exhaustive grid for parameters
Checkpoint selection	Configuration-bundle selection; the winning bundle is the artefact the system runs on
Regularisation	Schema constraints, evidence grounding, URI validation, and permissive filtering, which control fluent but unsupported output
Training duration, loss curves	Index build time and evaluation runtime; the parameter sweep curve is the equivalent diagnostic

Prompt refinement stops on three stated conditions rather than on visible improvement: field extraction is stable across repeated runs on the same input, no schema violations occur on the development split, and no new error class appears relative to the previous prompt version. Parameter selection stops when the sweep grid is exhausted.

3.2 Integrated architecture

The merged request path runs in three steps, all within one application and one database.

Step	Endpoint	Behaviour
1	Resume upload	The vision parser returns a validated candidate profile. Sensitive fields are encrypted at rest, the profile is stored, and a profile identifier is streamed back to the client
2	Recommendation	Given that identifier, the module reads and decrypts the stored profile through the parsing module's public service, translates the schema across the module boundary, embeds the flattened profile, retrieves the twenty closest occupations, re-ranks deterministically to five, and calls the explanation model. The result is persisted as a run record with provenance columns
3	Retrieval of results	Any module can read the stored recommendation run by profile identifier

Job Discovery and Matching consumes exactly what step three exposes — a candidate profile and a recommended occupation — so the remaining integration work is a read against an existing interface rather than a redesign.

The database holds parse job records, encrypted candidate profiles, and an audit trail from the first Alembic revision; recommendation runs with a cascading foreign key and provenance columns from the second; and 3,039 occupation vectors with an HNSW cosine index created by the setup script. Recording the embedding provider, explanation model, and skill-bonus weight on every run row is a deliberate reproducibility measure, because two of those three values changed during this milestone.

3.3 Final configuration bundle

Because no models are trained, there are no checkpoints to compare. What is fixed instead is the bundle below, which is the artefact the system currently runs on. Each value carries the basis on

which it was set: empirical, where it was selected by a measured sweep, or heuristic, where it was set by reasoning about the task and has not yet been varied and measured.

Module	Parameter	Value	Basis
All	Temperature	0	By design — reproducibility is required and sampling variety has no value in any task here
All	Embedding dimensions	768	Fixed by the encoder, identical across modules
Resume Parsing	Description-copying instruction	Verbatim-copy rule included	Empirical — two prompt versions compared, exact copy rate 20 to 45 percent
Resume Parsing	Routing and retries	Vision-first, 100-character text-layer probe, one page per call, one fallback retry	Heuristic — the working corpus contains no page with an extractable text layer
Career Rec.	Chunk contents	Preferred label, alternative labels, description, essential and optional skills	Empirical in part — optional skills added after finding them present for 2,857 of 3,039 occupations
Career Rec.	Vector distance	Cosine, explicit HNSW operator class	Empirical — measured against the previous default-distance index
Career Rec.	Skill-bonus weight	0.02	Empirical — eight values swept against MRR and Hit Rate
Career Rec.	Essential and optional skill weights	1.0 and 0.5	Heuristic — the ratio itself has not been swept
Career Rec.	Retrieval Top-K and Top-N	20 and 5	Heuristic — Hit Rate at 20 of 0.9186 confirms the window is wide enough, but alternative values were not compared
Job Matching	BM25 and embedding weights	0.5 and 0.5	Empirical — six configurations compared
Job Matching	Shortlist, hybrid and judge weights	15; 0.8 and 0.2	Heuristic — not varied in the sweep
Job Matching	Retrieved chunks per job at judge time	2	Heuristic — judge prompt size
Job Discovery	Search and crawl parameters	max_results 5, crawl limit 3, concurrency 3, timeout 8,000 ms	Empirical — five parameter sets compared
Job Discovery	Prompt variants	boolean_operators for query generation; comparative for the judge	Empirical — seven and six variants compared respectively

4. Evaluation Methodology

4.1 Protocol

Each measured component is evaluated offline against a fixed reference set, isolated from the stages around it. Career Recommendation is scored on retrieval and re-ranking only, because the explanation call is constrained to occupations the deterministic stages already selected and cannot change the ranking being measured. Job Matching is scored on a fixed candidate-job pool, so ranking quality is separated from crawl reliability. Job Discovery is scored on live queries precisely because search and crawl reliability is the thing being measured.

Every sweep ranks configurations by a weighted composite that blends the stage's quality metrics with a latency term, weighted at 15 percent in the ranking and search sweeps, with a hard latency ceiling applied as a pass condition in the prompt sweeps. Latency is included because all of these stages sit on an interactive path.

4.2 Ground truth

The resume gold set carries a resume category rather than an occupation label, so the Career Recommendation ground truth is a mapping: each of the 43 categories was mapped to a set of acceptable occupations, 134 mappings in total, generated by keyword matching against taxonomy titles and then reviewed by hand. A recommendation counts as correct when it matches any occupation in the set for that resume's category. The mapping is a proxy, and it is used to judge whether an acceptable occupation is retrieved at all and how highly it is ranked. Several categories map to overlapping occupation sets, so the ground truth supports rank-position measures rather than set-coverage measures, and only the former are reported for this module.

For Resume Parsing, 15 of the 86 gold records have been verified against their source images and 71 remain AI-drafted. Formal per-field Precision, Recall, and F1 were not run in this milestone, and that is recorded as a gap rather than filled with an estimate.

4.3 Cross-validation strategy

No k-fold cross-validation is used, and the reason is a property of the data. The gold set holds two resumes per category; a k-fold partition would place a single resume from a category in a fold, producing per-fold estimates too unstable to compare. The design used instead is a single stratified development and held-out split with near-duplicate groups confined to one side.

4.4 Baselines

Component	Baseline	Why this baseline
Career Recommendation re-ranking	Retrieval alone at Top-K 20, with re-ranking set to pass-through; and the previously specified rule that sorted on exact skill-match count with similarity only as a tiebreak	The first is the ordering the re-ranker is handed, so it is the floor it must beat; the second was the design of record that the current implementation replaced
Career Recommendation retrieval	The pre-migration index using the default distance function	Isolates the effect of the vector-store migration on retrieval quality
Job Matching ranking	Single-signal rankers, bm25_only and embedding_only, plus the previously configured 0.4 / 0.3 default	Tests whether the hybrid score beats either signal alone, and whether the configured weights were justified
Job Discovery, prompt stages	The previously configured default parameter set; the baseline prompt variant	Tests whether sweeping and prompt engineering contribute anything beyond what was already in place

4.5 Acceptance criteria and outcomes

Criterion	Outcome
Resume Parsing: per-field F1 at or above 0.75 on the held-out split	Not evaluated — the formal per-field run was not executed in this milestone
Resume Parsing: no schema violation on the development split, with a partial profile rather than a failed request	Met
Career Recommendation: re-ranking must not fall below retrieval alone on the primary measures	Met on Hit Rate at 1 and at 5. MRR sits 0.0005 below retrieval, which is truncation from twenty candidates to five rather than reordering damage
Career Recommendation: the correct occupation inside the initial retrieval window on the large majority of profiles	Met — Hit Rate at 20 is 0.9186
Career Recommendation: zero unrecoverable failures across the annotated set	Met — 86 of 86 profiles produced a ranked list
Job Matching: ranking quality at or above the previously configured default weights	Met — the selected weights match or exceed the default on the composite and are an order of magnitude faster
Job Discovery: search coverage sufficient to supply the ranking stage	Met — coverage is 1.0 in every configuration tested
Job Discovery: crawl success high enough to be treated as reliable	Not met — the best configuration reaches 0.50
Judge stage: schema-valid, latency-bounded, internally calibrated output	Not met — no prompt variant reaches a passing rate
Integration: Resume Parsing and Career Recommendation as one application over one database, verified on a real resume	Met
Integration: Job Discovery consuming the stored recommendation run	Not met — the module remains standalone

5. Performance Metrics

The three modules pose three different measurement problems, so no single metric family covers the system. Career Recommendation and Job Matching are ranking tasks and take retrieval and rank-correlation metrics. Job Discovery is a reliability problem rather than a model problem and takes success rates. The prompt-driven stages produce structured output and are measured on structural validity, internal consistency, and latency.

Metric	Definition	Why it is used here
Hit Rate at K	Fraction of profiles with at least one acceptable occupation in the top K	The user is shown a short list, so what matters is whether an acceptable occupation appears within the list they see
MRR	Mean reciprocal rank of the first acceptable occupation	Rewards putting an acceptable occupation near the top rather than merely somewhere in the list; the primary ordering measure

Metric	Definition	Why it is used here
Precision at 5	Fraction of the top 5 ranked jobs that are relevant	Job Matching only; matches what a candidate reviews first
NDCG at 10	Discounted cumulative gain against the ideal ordering, normalised	Rewards correct ordering across the ranked list rather than a binary cut
Spearman correlation	Rank correlation against the reference ordering	The most sensitive of the ranking measures on a ten-item pool, so the most likely to expose a genuine difference
Coverage score	Whether search returns enough unique candidate URLs for a query	The first place the discovery pipeline can fail
Crawl success rate	Fraction of attempted URLs yielding usable content	The second place it can fail, and in practice the binding one
Schema score	Fraction of calls returning schema-conformant JSON of the expected length and element types	Both prompt stages; a downstream parser depends on it
Diversity score	Token-overlap diversity across generated queries, required above 0.4	Query generation, where duplicate queries waste the search budget
Calibration score	Whether the categorical recommendation is numerically consistent with the returned interview probability	The judge, where an internally contradictory judgment is a failure even when the JSON parses
Latency and pass rate	A ceiling of 12 seconds for query generation and 20 seconds for the judge, applied as pass or fail	Both prompt stages sit on the interactive path

Two metrics used earlier in the project were dropped: Recall at 100, because the system never shows a candidate 100 occupations, and NDCG at 20 with Precision at 20, replaced by the at-10 and at-5 forms for the same reason.

5.1 Metrics deliberately not used

- **BLEU, ROUGE, BERTScore.** No stage generates free text against a reference. The explanation model justifies occupations already chosen deterministically, and the correct check on it is whether every occupation and skill it names exists in the retrieved evidence — enforced by URI validation and a constrained skill vocabulary rather than by n-gram overlap.
- **ROC-AUC and PR-AUC.** Both require a calibrated score against a binary label. The Career Recommendation match score is a deterministic decision-support value, explicitly not a probability, and the judge probability is not calibrated against any hiring outcome.
- **Confusion matrix.** There is no small fixed label set to confuse; ranking runs over 3,039 occupations. The informative equivalent is the per-category breakdown of where the correct occupation was missed, reported in the error analysis.
- **Hallucination rate.** Hallucination is prevented structurally rather than measured statistically: every occupation URI returned by the explanation model is validated against the retrieved set, and a non-matching entry is dropped. Across the annotated runs the guard never needed to fire.

6. Experimental Results

6.1 Resume Parsing

Resume Parsing runs as a working component inside the merged application. A resume is routed by the text-layer probe, rendered page by page where required, parsed against the schema, then

validated, merged across pages, normalised, deduplicated, encrypted, and stored. The fallback path and the partial-profile path both function. Aggregate per-field figures are not reported for this module, for the reason given in the methodology; the behavioural evidence available is reported in the integration verification below.

6.2 Career Recommendation

Evaluated against the 86-record annotated gold set on the migrated index.

Stage	MRR	Hit@1	Hit@3	Hit@5	Hit@10	Hit@20
Retrieval, Top-K 20	0.6110	0.4651	0.6977	0.8256	0.9070	0.9186
Re-ranked, Top-N 5	0.6105	0.4884	0.6860	0.8372	—	—
Held-out split, 52 resumes	0.6359	0.5192	0.7500	0.8269	—	—

Re-ranking lifts Hit Rate at 1 from 0.4651 to 0.4884 and Hit Rate at 5 from 0.8256 to 0.8372, holding MRR within 0.0005 of retrieval. The per-record diagnostic explains that near-tie: 3 records improved, 3 worsened, and 7 lost an acceptable occupation to the truncation from twenty candidates down to five. The module produced a ranked list for every profile — 86 of 86 evaluated, 0 failures.

On the held-out split the selected configuration matches retrieval at ranks 1 and 3, improves Hit Rate at 5 from 0.8077 to 0.8269, and holds MRR within 0.008 of retrieval. Those figures sit at or above the corresponding full-set figures, which is the pattern consistent with a configuration that is not overfitted to the set it was chosen on.

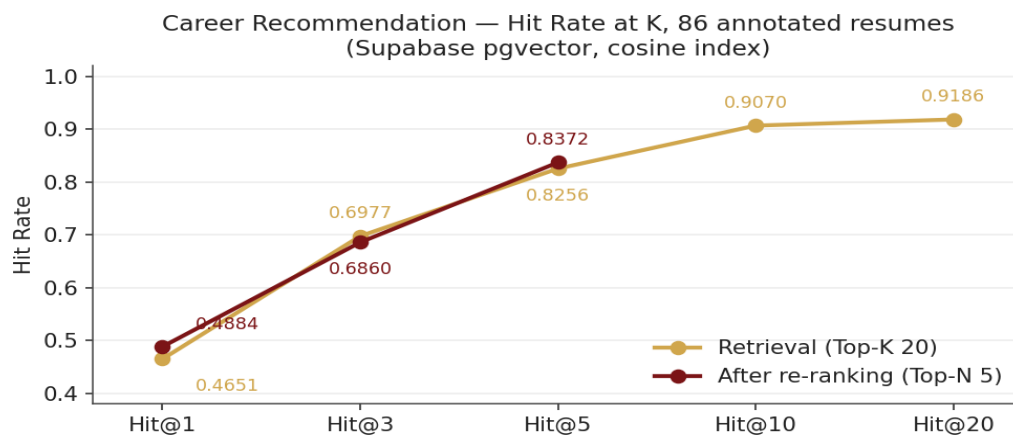


Figure 1. Hit Rate at K for retrieval and for the re-ranked output on the full annotated set. Re-ranking operates only within the top five, so it has no value at K of 10 or 20.

The migration to the hosted store also improved retrieval, because the previous index used the store's default distance while the new one uses an explicit cosine operator class. With chunk definition, encoder, and query construction unchanged, retrieval Hit Rate at 5 rose from 0.7791 to 0.8256.

6.3 Job Discovery and Matching

Four sweeps were run: two parameter sweeps on the stages that make no model call, and two prompt sweeps on the two stages that do.

6.3.1 Job Matching — ranking weight sweep, 86 candidates with 10 jobs each

Configuration	Composite	Precision@5	NDCG@10	Spearman	Latency (ms)	Pass rate
balanced_50_50 — selected	0.8977	0.8486	0.9795	0.8767	226	94.6%
embedding_heavy (0.3 / 0.7)	0.7670	0.8588	0.9793	0.8861	4,145	0%
bm25_only (1.0 / 0.0)	0.7656	0.8512	0.9793	0.8995	4,239	0%
current_default (0.4 / 0.3)	0.7593	0.8518	0.9791	0.8812	4,241	0%
embedding_only (0.0 / 1.0)	0.7518	0.8419	0.9706	0.8727	4,201	0%
bm25_heavy (0.7 / 0.3)	0.7503	0.8535	0.9789	0.8796	4,500	0%

Ranking quality is near-identical across all six configurations, and the separation is almost entirely latency — 226 milliseconds for the winner against more than 4,000 for every alternative. The practical conclusion is that on this data the split between the lexical and semantic weights is not a quality-sensitive parameter, and an equal split was selected.

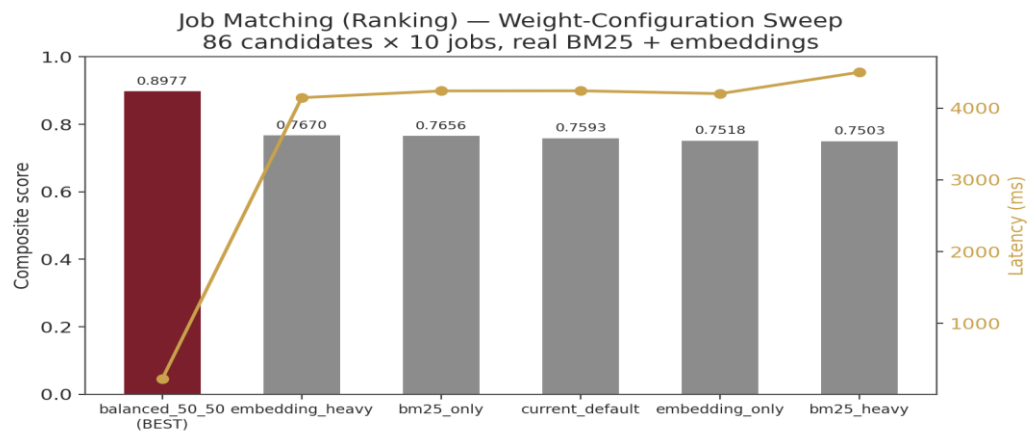


Figure 2. Composite score (bars) and latency (line) across the six ranking weight configurations. Ranking quality is flat; latency decides the winner.

6.3.2 Job Discovery — search and crawl sweep, five live queries

Configuration	Composite	Coverage	Crawl success	Latency (ms)
narrow_fast — selected	0.4500	1.0	0.50	2,703
high_concurrency	0.4152	1.0	0.45	6,868
current_default	0.4041	1.0	0.40	8,194
wide_net_slow	0.3618	1.0	0.067	16,030
low_concurrency_safe	0.3433	1.0	0.267	26,604

Coverage saturates at 1.0 everywhere, so the discriminating measures are crawl success and latency. The best configuration crawls successfully on half of attempted URLs; the widest succeeds on under 7 percent while taking roughly six times as long. Crawl success is reported as an open reliability problem rather than as a solved component.

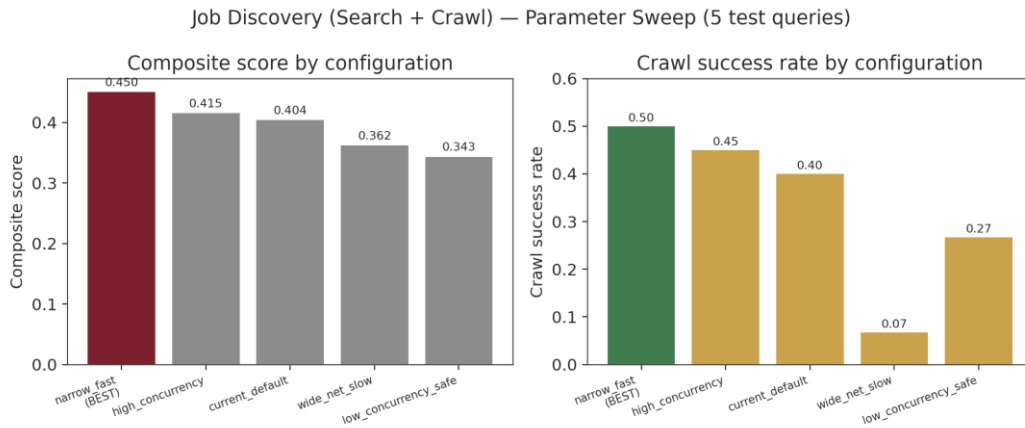


Figure 3. Composite score and crawl success rate across the five search-and-crawl configurations. Coverage is saturated at 1.0 everywhere and is not plotted separately.

6.3.3 Prompt sweeps

Query generation was tested across seven variants on four candidate profiles, and the batched judge across six variants on two job batches — 40 live model calls in total.

Stage and variant	Composite	Schema	Diversity or calibration	Latency (ms)	Pass rate
Query generation — boolean_operators, selected	0.5102	1.0000	0.8010	19	100%
Query generation — remaining six variants	0.3602 to 0.4306	1.0000	0.8010	26 to 33	100%
Judge — comparative, best available	0.2750	0.5000	0.5000	21,572	0%
Judge — narrative	0.2060	0.5000	0.5000	28,082	0%
Judge — baseline, weighted_criteria, strict_rubric, chain_of_thought	-0.0668 to -0.1500	0.0000	0.0000	27,872 to 35,716	0%

All seven query-generation variants return schema-valid and sufficiently diverse output, so the winner is decided on latency alone; the winning variant instructs the model to use search-engine operators mixed with plain-text queries and shows two worked examples of that pattern. The judge sweep is a negative result: no variant achieves a passing rate, and the best reaches only 0.5 on schema validity and 0.5 on calibration. It was selected because its rank-then-score instruction yields internally consistent probability values when it does return clean JSON, which every other variant fails to do at all. It is carried in the current limitations rather than in the results.

6.4 Integration verification

The merged path was exercised end to end on a real image-only resume drawn from the corpus, to confirm that the handoff between the two modules works on production-format input rather than on a hand-written test object. The chain ran without manual intervention: the parser produced a schema-valid profile, the profile identifier was returned over the event stream, the recommendation module read and decrypted that stored profile through the parsing module's public service, translated the schema across the module boundary, embedded, retrieved, re-ranked, explained, and persisted the run.

Stage	Result
Parsing	Valid on the first attempt: fallback not invoked, no review flags raised, descriptions copied verbatim, and personal identifiers correctly excluded
Normalisation	Six of seven skill bullets retained, the seventh correctly removed as a near-duplicate; sections absent from the source returned empty rather than populated by inference
Recommendation	Status ok. Top-5: agricultural engineer, agricultural equipment design engineer, test engineer, agricultural machinery technician, agronomist
Ranking accuracy	The candidate's actual occupation returned at rank 1, with four of the five recommendations in the correct domain
Confidence handling	The similarity-only path engaged and reported low confidence across the set, with a clear explanation to the user

The confidence behaviour is the most useful part of this verification. This resume writes its skills as full sentences, so the deterministic skill-overlap signal had nothing to match against the taxonomy's short noun-phrase labels. The system detected that condition, ranked on semantic evidence alone, returned the correct occupation at the top of the list, and reported low confidence with an explicit statement of why — exactly the designed behaviour for an input where the supporting evidence is thin. It demonstrates the safeguard working on real input, alongside the aggregate quality figures measured over the full 86-resume set above.

7. Baseline Comparison

Every selected configuration is compared against something that already worked, so that each component earns its place rather than being assumed to help.

7.1 Career Recommendation

Two baselines apply: retrieval alone, which is the ordering the re-ranker is handed, and the rule specified earlier in the project, which sorted on exact skill-match count with similarity only as a tiebreak. Measured on the held-out split, the improvement over the previously specified rule is large — MRR rises from 0.4949 to 0.6359 and Hit Rate at 5 from 0.6731 to 0.8269. Against retrieval alone the change is small and mixed: Hit Rate at 5 improves by 0.0192 while MRR falls by 0.0078. The clearest value of the current step is that it is a correct, inspectable, deterministic stage that no longer degrades the retrieval ordering, which the earlier rule did.

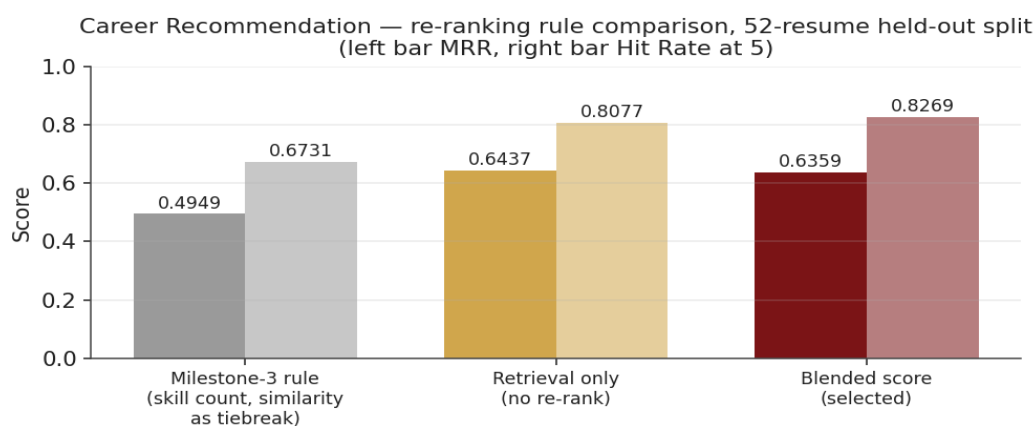


Figure 4. Re-ranking rule comparison on the held-out split. The previously specified rule sits well below the retrieval ordering it was meant to improve; the blended score sits at or above it.

7.2 Job Matching, Job Discovery, and the prompt stages

The hybrid ranking score does not beat either single signal on ranking quality — Spearman is highest for the lexical signal alone at 0.8995 against 0.8767 for the selected configuration, and Precision at 5 is highest for the embedding-heavy variant. The composite gap in favour of the selected configuration comes almost entirely from an eighteen-fold latency difference, so the improvement claimed here is an efficiency improvement rather than a quality one.

Job Discovery is the one sweep where the selected configuration improves on the configured default on both dimensions, raising crawl success from 0.40 to 0.50 while cutting latency by 67 percent. For the prompt stages, query generation was already saturated on correctness at the baseline, so prompt engineering bought latency; for the judge it moved schema validity from 0.0000 to 0.5000 without moving the pass rate off zero.

Component	Baseline	Selected	Improvement and nature
Career Recommendation re-ranking	MRR 0.4949, previously specified rule	MRR 0.6359	+0.1410, quality
Career Recommendation retrieval	Hit@5 0.7791, default-distance index	Hit@5 0.8256, cosine index	+0.0465, quality
Job Matching ranking	Composite 0.7593 at the configured default	Composite 0.8977	+0.1384, efficiency; ranking quality unchanged
Job Discovery search and crawl	Crawl 0.40 at 8,194 ms	Crawl 0.50 at 2,703 ms	+0.10 and –5,491 ms, both
Query generation prompt	Composite 0.3646	Composite 0.5102	+0.1456, efficiency; correctness already saturated
Batched judge prompt	Composite -0.0668, schema 0.0000	Composite 0.2750, schema 0.5000	Partial; structural validity only, stage still unreliable

8. Configuration Parameter Analysis

There are no hyperparameters in the training sense. The tunable surface is prompt text, output schemas, retrieval parameters, and scoring weights, and each sweep is an exhaustive grid over a small discrete set of values with everything else held constant. Grid search is appropriate here because each parameter has a handful of meaningful values and a run is cheap on the non-model stages, so an exhaustive grid gives the full shape of the response curve rather than a single optimum. In two of the four sweeps the shape mattered more than the optimum.

8.1 Career Recommendation — the skill-bonus weight

The re-ranking step blends cosine similarity from retrieval with exact overlap between the candidate's skills and the occupation's taxonomy skill list. The sweep answers how much influence the overlap signal should carry, parameterised as a single weight and swept across eight values from a pass-through to skill-dominant settings.

Skill-bonus weight	0.00	0.02	0.05	0.08	0.10	0.15	0.25	0.40
MRR	0.5849	0.5965	0.5866	0.5457	0.5328	0.4973	0.4866	0.4843
Hit@1	0.4651	0.4884	0.4767	0.4070	0.3953	0.3372	0.3256	0.3256
Hit@5	0.7791	0.7907	0.7907	0.7907	0.7674	0.7674	0.7674	0.7558

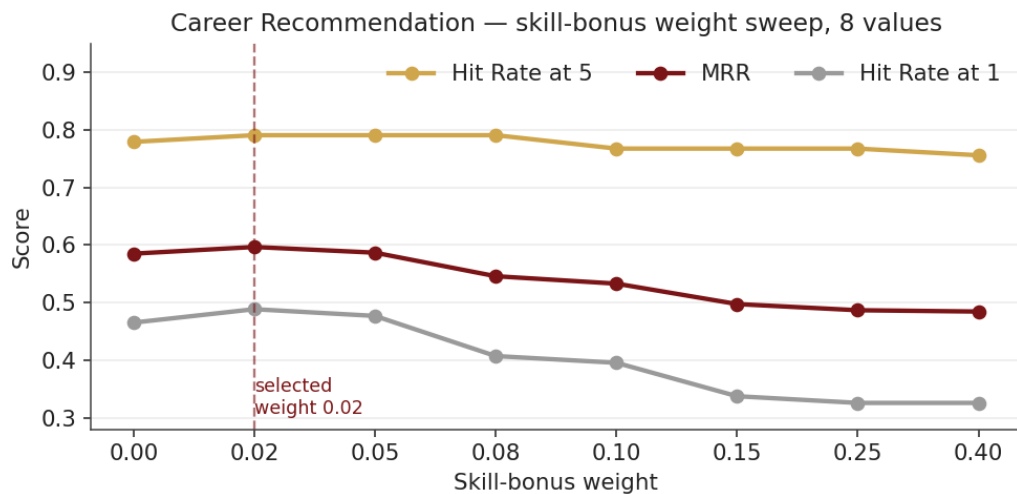


Figure 5. Response of MRR, Hit Rate at 1, and Hit Rate at 5 to the skill-bonus weight. The monotonic decline above 0.05 is the substantive finding of this sweep.

At weight 0.00 the step is a pass-through and reproduces the retrieval ordering exactly, which confirms the implementation is correct and establishes the floor. Weight 0.02 improves on that floor on MRR, Hit Rate at 1, and Hit Rate at 5 and was selected. Above 0.05 every measure declines monotonically.

The shape of that curve is the substantive result rather than the winning value. Exact skill matching contributes usefully only at low weight, as a tiebreaker between occupations of comparable similarity, and not as a primary ranking signal — resume skill text and taxonomy skill labels draw on different vocabularies, so exact string overlap fires infrequently, and when it does fire it is often on a generic term. A skill-dominant setting therefore lets one incidental match outrank an occupation with strong semantic similarity and no exact match. These figures were measured on the pre-migration index; the selected weight was carried across unchanged and performs slightly better on the migrated store.

8.2 The remaining sweeps

For the Job Matching weights the response is flat on every quality metric and steep on latency, across a space that includes both single-signal extremes, so the conclusion drawn is not that an equal split is optimal but that the parameter is not quality-sensitive on this evaluation pool. For search and crawl, the response is monotonic in an unexpected direction — the narrower the search, the higher the crawl success — because a wider net surfaces URLs that are disproportionately resistant to extraction, so there is no trade-off to balance and the narrowest configuration wins outright. For the judge prompt the response is bimodal: two variants reach 0.5 on schema and calibration, four reach zero, and none passes.

8.3 Rejected configurations

Rejected	Measured outcome	Reason
Skill-bonus weights 0.05 to 0.40	MRR falls from 0.5866 to 0.4843	Monotonic degradation; overlap is a tiebreaker, not a ranking signal
Previously specified re-ranking rule	MRR 0.4949, Hit@5 0.6731 held-out	Sits below the retrieval ordering it was meant to improve
Five alternative ranking weight sets	Composite 0.7503 to 0.7670 against 0.8977	Latency above 4,000 ms with no compensating quality gain

Rejected	Measured outcome	Reason
Four alternative search and crawl sets	Crawl 0.067 to 0.45; latency 6,868 to 26,604 ms	Lower crawl success and higher latency than the narrow configuration
Six alternative query prompts	Identical schema and diversity; 26 to 33 ms	Slower with no quality difference
Five alternative judge prompts	Schema 0.0000 to 0.5000; pass rate 0%	None passes; the retained variant is least unreliable rather than acceptable
Hosted embedding model for the occupation index	Build failed at 100 of 3,039 documents	Free-tier quota of roughly 100 requests per minute and 1,000 per day; paid access was not available

9. Ablation Study

Several swept configurations correspond to removing a component entirely, so the sweeps double as ablations. Those cases are collected here and read as component-contribution evidence.

Component removed	Measure	With	Without	Contribution
Semantic similarity as primary signal	MRR, 86 resumes	0.5965	0.4843	+0.1122
Cosine distance function	Retrieval Hit@5, 86 resumes	0.8256	0.7791	+0.0465
Deterministic re-ranking step	Hit@5, held-out 52	0.8269	0.8077	+0.0192
Skill-overlap bonus at weight 0.02	MRR, 86 resumes	0.5965	0.5849	+0.0116
Lexical signal in Job Matching	Precision@5	0.8486	0.8419	+0.0067, within noise
Semantic signal in Job Matching	Precision@5	0.8486	0.8512	–0.0026, within noise
Narrow search constraint	Crawl success	0.50	0.067	+0.433
Similarity-only confidence path	Behavioural	Correct occupation at rank 1, low confidence, explicit caveat	Unsupported confidence or no result	Qualitative

Three readings follow. The semantic signal is carrying Career Recommendation — driving the skill weight to 0.40 approximates removing similarity as the primary signal and costs 0.1122 on MRR, the largest single component effect measured anywhere in the system. The distance function, a one-line index parameter, contributes more than the entire deterministic re-ranking stage. And the skill-overlap bonus is small, positive, and worth keeping only at low weight.

The Job Matching ablation is a negative result and is reported as such: neither signal can be shown to be necessary on this evaluation pool. The hybrid score is retained because the reasoning behind it remains sound — the lexical component catches exact tool and certification names that embeddings

blur, the semantic component catches equivalent skills phrased differently — but that reasoning is not currently supported by measurement, and the pool is very likely too small to test it. The judge stage cannot be ablated cleanly, because it contributes 0.2 of the final score but does not yet produce reliably parseable output.

For Job Discovery, removing the constraint on search breadth drops crawl success from 0.50 to 0.067 and raises latency from 2,703 to 16,030 milliseconds, with coverage unchanged. Breadth contributes nothing and costs a great deal, which is the opposite of the intuition that motivated a wider net.

10. Error Analysis

10.1 Career Recommendation

Our analysis identified four distinct root causes for why the system failed to return an acceptable occupation in the top five recommendations:

- **The Vocabulary Gap:** The dominant cause of error is the mismatch between the language candidates use on their resumes and the standardized labels used in the ESCO taxonomy. Because these vocabularies differ, relying heavily on exact skill-matching degrades the system's accuracy.
- **Domain-Specific Blind Spots:** Three resume categories consistently failed for entirely different reasons:
 - ◆ **Arts:** The category is broad and resumes describe artistic practice, while the taxonomy expects narrow, specific job titles.
 - ◆ **Management:** Management vocabulary is so generic that it overlaps with a massive portion of the 3,039 occupations, making it difficult for the embedding to find a specific direction.
 - ◆ **Network Security Engineer:** This is a modern job title that simply does not exist in the taxonomy. This is a database coverage failure, not a retrieval failure.
- **List Truncation:** The system retrieves 20 candidates but only shows the top 5 to the user. On 7 of the 86 test records, the correct occupation was successfully found but was lost when the list was truncated to 5.
- **Encoder Token Limits:** The local embedding model has a maximum context window of 512 tokens. Because the longest occupation documents in the database reach roughly 1,300 tokens, the tail ends of those documents are cut off and never embedded into the vector store.

10.2 Job Discovery and Matching

Discovery-stage failures are on the source side rather than in the pipeline. A representative run found five unique URLs, attempted three, and extracted none, at 13.5 seconds; the pipeline did not fail, it skipped the failed URLs and completed. Postings that render client-side, block automated access, or omit structured metadata leave the deliberately zero-model extraction stage with nothing to work from. The sweep supports this reading: success falls as breadth rises, so the additional URLs a wider search surfaces are disproportionately the ones that resist extraction, while coverage stays at 1.0 throughout.

The judge stage fails in two distinct ways under the same prompt at temperature 0. The baseline variant returned text that was not JSON at all. The selected variant failed one of its two cases by prepending visible reasoning before the JSON array, breaking the parser, and the other by returning valid output at 178,664 milliseconds against a 20,000 millisecond ceiling. Two different faults under one prompt suggest variable model behaviour on a task at the edge of what this judge model does reliably — batch scoring five items with numerically consistent probabilities in a fixed structure — rather than a single fixable bug. Because all six variants reach a 0 percent pass rate, further prompt rewording is not the remedy; the plausible remedies are a different judge model or a stricter output-constraining mechanism.

One diagnostic finding is worth separating from the results. Ranking quality is statistically flat across all six weight configurations, varying only in the second decimal place across a space that includes both single-signal extremes. That is unusual for a weight sweep and is more likely a property of the

evaluation set — ten jobs per candidate from a curated dataset — than of the ranker, so the near-tie should not be read as evidence that ranking weights do not matter on larger, messier pools.

10.3 Defects found during integration

The most consequential defect produced no error at any layer. During the migration, the vector-store client library's insertion helper wrote malformed embeddings: content and metadata landed correctly, the insert returned success, and row counts were right. The corruption was detectable only by measuring the vectors themselves — a stored vector had an L2 norm of 0.5951 where this encoder always produces approximately 1.0, and comparing a stored vector against a fresh embedding of its own document text gave a cosine similarity of -0.0072 , so a document was orthogonal to itself. The user-visible symptom was retrieval nonsense, with a data-analyst query returning occupations such as derrickhand and pizzaiolo at similarity around 0.06.

The fix was to bypass the helper and write the vector literal directly through the database driver. The more durable change is the second one: a verification step now ships inside the ingestion script, so a rebuilt index is checked for vector norm and self-similarity as part of the build rather than trusted because the insert succeeded.

Defect	Symptom	Root cause and resolution
Search-function signature mismatch	Every retrieval call returned PGRST202	The client sends only the query embedding and an optional filter and applies the result limit client-side, while the initial SQL function required an explicit count argument. The function was recreated to match the caller's contract
Silent row-limit cap	A check against the 3,039-row document table returned 1,000 rows	The REST layer applies a default row limit. The check would have concluded the table was under-populated and re-ingested roughly 2,000 duplicates. Caught before the re-run; the check now paginates
Vendor model retirement, three occurrences	A configured identifier returns 404 for new keys; a second is unavailable; a hosted embedding endpoint is quota-blocked	Provider-side deprecation and quota policy. Mitigated by holding every model identifier in environment configuration, so a substitution is a configuration change rather than a code change

11. Robustness

11.1 Generalisation to unseen data

The held-out comparison is the primary generalisation evidence: the configuration selected on the annotated set scored at or above its full-set figures on every measure when run on the 52 unseen resumes. For Job Matching the equivalent evidence is stability — ranking quality holds across the full 86-candidate golden set and across six materially different weight configurations, with no configuration collapsing.

Three limits on generalisation are visible in the evidence. The resume corpus is entirely single-page and image-only, so the text path and multi-page merge are less exercised than the vision path. The occupation taxonomy is European in origin, so regional and emerging titles map less cleanly, which is directly visible in the network-security category. And the Job Matching pool is small and curated.

11.2 Fail-soft behaviour

Condition	Behaviour
Parser output fails schema validation or the completeness check	One fallback retry against a second model, then a partial profile with review flags rather than nothing
Candidate skills produce no overlap with any retrieved occupation	Similarity-only ranking, confidence reported as low, with an explicit message to the user
The explanation model is unavailable, or no candidates survive retrieval	An explicit degraded or no-candidates status with the deterministic ranking intact, rather than a failed request
The explanation model names an occupation outside the retrieved set	The entry is dropped by URI validation before the response is returned
A job posting URL fails to crawl, or all queries return nothing	The URL is skipped and the run continues; an empty shortlist carries a no-matches status rather than raising

Across the annotated evaluation the result of this design is that 86 of 86 profiles produced a ranked list and none failed.

11.3 Noise tolerance and adversarial input

A separate robustness set was generated from the resume corpus using small rotations, moderate Gaussian blur, JPEG compression, contrast and resolution reduction, and light image noise, with each augmented document inheriting the annotation and split of its source so a transformed copy cannot cross the evaluation boundary. Positive candidate-job pairs are augmented through controlled skill-synonym replacement from a manually reviewed alias list, applied after splitting and only within the development split. Augmentation is used for robustness testing only, never to expand a training set, since there is none.

Both text-bearing modules treat their input document as untrusted data: resume text and job descriptions are clearly delimited, cannot alter the system instruction, and the output schema is a hard gate regardless of content. This matters more here than in a typical extraction pipeline, because both inputs are authored by an interested party. Two structural controls limit the effect of a fluent but unsupported model output — the explanation model may only discuss occupations present in the retrieved set and may only name skill gaps from that occupation's own skill list, and every returned URI is validated before the response leaves the module. No formal adversarial test suite was executed, so these are implemented controls rather than measured defences.

12. Computational Performance

Stage	Measured	Notes
Resume parsing, per resume	96.0 seconds average	Five-resume sample with the verbatim-copy prompt; one hosted call per page
Career Recommendation, per profile	1.5 to 1.8 seconds	From 132 to 158 seconds across 86 profiles against the hosted store
Job Matching ranking	226 ms	Against more than 4,000 ms for every rejected configuration

Stage	Measured	Notes
Job Discovery search and crawl	2,703 ms	Against 26,604 ms for the slowest configuration tested
Query generation	19 ms	Range 19 to 33 ms across seven variants
Batched judge	21,572 ms median	Exceeds the 20-second ceiling; one case reached 178,664 ms
Occupation index build	Minutes on CPU	One offline pass over 3,039 documents with the local encoder

Centralising the vector store raised the Career Recommendation evaluation from 11.7 seconds to 132 to 158 seconds for the same 86 profiles. The cause is network round-trip time to the hosted region rather than additional computation, and it is a per-query cost rather than a scaling problem, since the index is queried once per profile regardless of its size. It is recorded here as a real cost of the migration alongside the retrieval-quality gain the same change produced.

Two figures in the table are open defects rather than acceptable measurements: the judge exceeds its own latency ceiling at the median, and 96 seconds per resume is slow enough to shape the experience of the whole pipeline. On the second, an alternative hosted model averaged 14.3 seconds on the same five-resume sample, but that comparison covers only latency and exact copy rate on five documents, not schema validity or fallback behaviour, so no configuration change was made on that evidence.

Resource	Footprint
GPU	Zero, everywhere, by design. All large-model work is hosted and the only local model is a 110M-parameter CPU encoder
Memory	Approximately 600 MB to 1 GB for the application stack, plus 100 to 200 MB per concurrent headless browser context in the crawler
Stored vectors	3,039 vectors of 768 dimensions, roughly 9 MB of raw float data plus index structure and document text
Database storage	Roughly 0.3 to 0.5 GB available on the free tier; the job store is deduplicated by a hash of the normalised posting URL, so storage grows with distinct postings rather than with searches
Call budget	One call per resume page plus at most one fallback; one local embedding and one explanation call per recommendation; exactly two model calls per Job Discovery run regardless of job count

The binding cost constraint on this project was never money but free-tier quota, and that constraint is what forced the local encoder, the offline index build, the batched judge, and the deterministic extraction stage.

13. Limitations

13.1 Dataset and methodology

- The Career Recommendation ground truth is a keyword-generated, hand-reviewed proxy rather than direct occupation labels, and several categories map to overlapping occupation sets, so a recommendation is judged on whether an acceptable occupation is retrieved and how highly it is ranked.
- Only 15 of the 86 resume gold records have been verified against source images; the remaining 71 were AI-drafted. Projects appear in 5 records and certifications in 27, so per-field results for those fields would rest on very few examples.

- The skill-bonus weight sweep was executed over all 86 records including the 52 held out, so for that one parameter the separation between selection and confirmation is not strict, and the sweep has not been repeated on the migrated store. The held-out comparison for the selected value was run separately and is clean.
- Resume Parsing has no formal per-field result, so its stated acceptance criterion is unevaluated rather than met or missed.
- Prompt sweeps depend on live hosted models, so reproducibility holds only while a given model version remains available; no formal adversarial test suite was executed.

13.2 Model and system

- The batched judge has no passing configuration across six variants; as deployed, this stage is an unresolved reliability gap rather than a working component.
- Crawl success reaches 0.50 on the best configuration, so roughly half of discovered postings cannot currently be extracted.
- The local encoder truncates at 512 tokens against occupation documents reaching roughly 1,300, so the longest documents are not embedded in full.
- Several values remain heuristic and are labelled as such in the configuration bundle: retrieval Top K and Top-N, shortlist size, the hybrid and judge weights, and the essential and optional skill weights.
- The re-ranking step's contribution over retrieval alone is small, and the hybrid ranking score cannot be shown to beat either single signal on the current pool.

13.3 Engineering and operational

- Job Discovery and Matching is not integrated with the shared profile and database flow; route prefixes are inconsistent across modules; and the central configuration module still carries retired model identifiers as its default values, so a deployment without environment overrides would start against models that no longer exist.
- Development API credentials circulated through team debugging sessions during this milestone. They are development-scope keys held in environment configuration, not production credentials, and are treated as needing replacement before any shared deployment.

13.4 Bias and ethical considerations

- **Taxonomy and corpus bias.** A European occupational taxonomy carries the labour-market structure of the region that produced it, so candidates whose roles are common elsewhere receive weaker matches — measurable in the per-category results rather than hypothetical. The resume corpus over-represents some categories by a factor of roughly twelve, which the gold set counters by sampling two resumes per category.
- **Decision support, not decision making.** The match score is a deterministic decision-support value and explicitly not a probability, and the judge's interview probability is not calibrated against real hiring data. Presenting either as a prediction of employment outcome would be unsupported by anything measured here.
- **Personal data.** Email addresses and telephone numbers are never extracted or persisted. Retained sensitive fields are encrypted at field level and decrypted only for the authenticated owner, original uploads are never written to persistent storage, rendered page images are deleted immediately after parsing, and every profile access is written to a separate audit log.

14. Possible Improvements

Each option below follows from a specific measurement in this report and is stated with what it would cost rather than as a commitment.

- **Closing the vocabulary gap.** Embedding the skill labels and matching them semantically rather than by string equality would let the overlap signal fire on paraphrases, at the cost of an additional embedding pass over 13,939 labels. A curated alias table would be cheaper and more

inspectable but needs maintenance. A cross-encoder re-ranker over the top twenty candidates would be the strongest and the most expensive, adding a model pass per query.

- **Encoder truncation.** Splitting the longest occupation documents into multiple vectors under one occupation identifier, or moving to a longer-context encoder, would remove the truncation. Since the effect has not been isolated, the first step this suggests is a measurement on the affected subset rather than a change.
- **The judge stage.** Six instruction styles at a 0 percent pass rate is enough evidence that prompt engineering is not the remedy. The structural options are constrained decoding that makes non-JSON output impossible, a different judge model, or unbatching at the cost of the call-budget discipline. A fourth is to treat the stage as optional and let the deterministic hybrid score stand alone when it fails, which the 0.8 to 0.2 final weighting already makes viable.
- **Crawl reliability.** Because success falls as breadth rises, targeting beats broadening restricting search to domains with a measured extraction success rate, tracked per domain in the job store, would concentrate effort where it works. A secondary extraction path for postings without structured metadata would raise coverage but reintroduce a model call per posting.
- **Strengthening the evaluation.** Re-running the weight sweep on the development split alone and on the migrated index would make the separation strict; enlarging the Job Matching pool beyond ten jobs per candidate would test whether the flat weight response belongs to the ranker or the pool; and sweeping the values currently labelled heuristic would convert them from reasoned choices into measured ones.
- **Completing the integration.** Job Discovery consumes a candidate profile and a recommended occupation, both already persisted as a run keyed by profile identifier, so the work is a read against an existing interface. Aligning route prefixes and replacing the retired default model identifiers are small, independent items in the same area.

15. Conclusion

This milestone merged two of the three modules into a single application over a centralised database and measured the system with retrieval, ranking, and reliability metrics against explicit baselines. The merged path runs end to end on production-format resumes. The vector-store migration improved retrieval Hit Rate at 5 from 0.7791 to 0.8256 while costing an order of magnitude in evaluation runtime through network latency.

Component	Headline result	Status
Resume Parsing	Working end to end within the merged application; no formal per-field metrics	Functional, unmeasured
Career Recommendation	MRR 0.6105, Hit@1 0.4884, Hit@5 0.8372, Hit@20 0.9186, 0 failures in 86	Measured
Career Recommendation, held out	MRR 0.6359, Hit@5 0.8269 on 52 unseen resumes	Measured, generalises
Job Matching ranking	Precision@5 0.8486, NDCG@10 0.9795, Spearman 0.8767 at 226 ms	Measured at module level
Job Discovery search and crawl	Coverage 1.0, crawl success 0.50, latency 2,703 ms	Measured; reliability below acceptance
Query generation	Schema 1.0000, diversity 0.8010, pass rate 100%	Measured, passing
Batched judge	Schema 0.5000, calibration 0.5000, pass rate 0%	Measured, failing



Component	Headline result	Status
Integration	Two of three modules merged over one database and one application	Partially complete

The Resume Parsing and Career Recommendation path is ready to be used as decision support and is not ready to be used as an automated decision, which matches how it is presented to the user: a short, ranked list with confidence bands and an explanation grounded in retrieved evidence. Job Discovery and Matching is ready in neither sense; its two dependencies on the outside world, reading job pages and obtaining reliable, structured output from a judge model, do not currently meet the criteria set for them, and both are reported here with the measurements that establish them.

The summary of the system as a whole is that its measured components are measured properly, its unmeasured component is identified as unmeasured, and its failing component is reported as failing. That is the state the evaluation was run to establish.

16. Team Consent

Name	Roll Number	Initial
Gaurav Kumar	22f1001105	GK
Dev Gupta	22f2000888	DG
Abhinav Ohri	24f1002064	AO
Pranav N	22f2000117	PN
Praveena N	22f3001454	PN